Ravani Roshan

Backend Developer · AI Infrastructure Engineer · LLM Systems Builder
Ahmedabad, Gujarat, India · +91 91060 95375 · ravaniroshansingh@gmail.com · github.com/RavaniRoshan

Professional Summary

Computer Science Engineering undergraduate focused on backend development for AI systems, API infrastructure, agent tooling, and production-ready developer platforms. Built backend-heavy AI infrastructure including a hermetic multi-agent coding system in Rust, a 59-tool MCP automation server, and an in-process AI SDK guardrail layer. Strong experience with Python, Rust, Node.js, FastAPI, PostgreSQL, Docker, cloud deployment, and LLM API integration.

Education

Silver Oak University — Ahmedabad, Gujarat, India
B.Tech, Computer Science Engineering — *CGPA: 8.0*

Experience

Google Developers Group — Oct 2023 – Dec 2025
Core Developer — *Ahmedabad, Gujarat, India*

- Led engineering sessions for a 200+ developer community on backend architecture, API integration, AI tooling, and deployment workflows.
- Mentored peers on building maintainable software using Python, TypeScript, databases, cloud platforms, and LLM APIs.
- Contributed to community projects across web infrastructure, applied ML, automation tooling, and AI developer platforms.

Projects

Niki – Hermetic Multi-Agent Coding System | *Rust, Docker, CLI, Multi-agent* — Jul 2026 – Present

- Built a hermetic multi-agent coding system in Rust where isolated AI agents independently plan, code, test, and review inside Docker sandboxes, producing reviewable git branches.
- Designed process isolation architecture with BYOK LLM integration, per-agent resource limits, and deterministic sandbox teardown.

WinScript MCP | *Python, MCP, FastMCP, SQLite, Docker, Windows Automation* — Apr 2026 – Present

- Built a Python-based MCP server exposing 59 tools for Windows app control, UI interaction, Office COM automation, filesystem ops, clipboard, screenshots, and audit logging.
- Unified multiple automation layers (UI Automation, COM, Win32, OCR fallback, SQLite storage) behind agent-callable backend APIs.
- Designed state-aware responses returning action results, active-window changes, selector confidence, and failure context for reliable AI-agent execution.

backstop – In-Process AI SDK Guardrails | *Python, OpenAI/Anthropic SDK, httpx* — Jul 2026 – Present

- Built an in-process AI SDK safety layer enforcing token budgets, circuit-breaking, retry policies, and real-time metrics for OpenAI/Anthropic clients at transport level.
- Implemented AIMD congestion control, priority-based admission, gzip-cached replay, and a multi-agent diff CLI (Wedge) for parallel agent cost isolation.

REI – Repository Evolution Intelligence | *Python, AST Parsing, Git Mining, MkDocs* — Jun 2026 – Present

- Built a software-intelligence framework that predicts file impact from source-code changes using static dependency analysis and historical change coupling.
- Structured as a reusable Python package with MkDocs project site.

Technical Skills

Backend: Python, Rust, FastAPI, Node.js, Express, REST APIs, WebSockets, serverless functions, API design
AI Infrastructure: MCP servers, LLM API integration, LangChain, LangGraph, agent tool-calling, RAG pipelines, AI observability
Databases: PostgreSQL, MySQL, Supabase, SQLite, Redis/Vercel KV
Cloud & DevOps: Docker, Docker Compose, Vercel, Railway, Fly.io, AWS, GCP, Nginx, GitHub Actions, Linux/Windows
Frontend: JavaScript, TypeScript, React, Next.js, Tailwind CSS, shadcn/ui, dashboards

Certifications

Google: MLOps for Generative AI (Dec 2025), Responsible AI: Privacy & Safety (Jan 2026)
DeepLearning.AI: LangChain for LLM Application Development, AI Agents in LangGraph
Anthropic: Claude Code in Action (Mar 2026)
Other: Foundational C# with Microsoft (freeCodeCamp, Nov 2024)